

RESEARCH ARTICLE

Large-scale field studies inform adaptive management of California wetland restoration

Kathryn M. Beheshti^{1,2} , Stephen C. Schroeter¹, Andres A. Deza¹, Daniel C. Reed¹, Rachel S. Smith¹, Henry M. Page¹

Salt marsh vegetation provides the structure that supports key wetland functions and the success of wetland restoration efforts depend on the establishment of vegetation, which can take decades. Field experiments aimed at identifying the factors that limit the colonization rate of marsh vegetation can be useful for guiding restoration efforts. We used field experiments and additional measurements to determine the causes for persistently low vegetation cover in a restored salt marsh in southern California. First, we characterized the soil properties of the marsh and found that high salinity and low soil moisture increased with elevation and improved with vegetation. Using this information, we conducted two field experiments—one in the high marsh and the other in the mid marsh to determine the effectiveness of different remedial actions (e.g. irrigation, soil decompaction, soil amendments, and planting and seeding different species) on increasing vegetation cover; the mid marsh experiment evaluated planting and seeding only. In the high marsh, plantings increased cover, and irrigation in combination with soil decompaction facilitated natural plant establishment. In the mid marsh, increases in vegetation cover varied by species planted. In both experiments, the marsh dominant (*Salicornia pacifica*) naturally recruited, increasing cover whereas seeding did not increase cover. In general, the magnitude and pace of vegetation growth was greater in the mid marsh relative to the high marsh that received less tidal inundation. Collectively, our results show how information gained from experiments conducted during wetland restoration can inform the processes underlying the establishment of vegetation.

Key words: adaptive management, decompaction, irrigation, planting, soil properties, wetland restoration

Implications for Practice

- To increase vegetation cover in wetland restoration, practitioners should consider how hydrologic and soil conditions change with elevation.
- In arid, drought affected wetlands, the potential amelioration of common physical stressors (high soil salinity, low soil moisture content, high bulk density) by early marsh plant colonizers may kickstart natural recruitment, seedling establishment and growth relative to nonvegetated areas.
- Field experiments are valuable in guiding best practices (e.g. irrigation and soil decompaction) for active salt marsh restoration by identifying likely stressors that may inhibit vegetation establishment and growth.

Introduction

Coastal wetlands are considered a globally threatened habitat, largely due to human impacts (Kennish 2002; Zedler & Kercher 2005; Davidson 2014). For decades tidal wetlands were diked, drained, and filled to accommodate coastal development. Watersheds that supply the essential fresh and tidal waters to wetlands have been permanently reconfigured to accommodate municipal freshwater demands and maintain or expand navigable waters in harbors and ports (Van Dyke & Wasson 2005). Run-off has transformed many once nutrient-limited wetlands

into eutrophic systems (Merrifield et al. 2011; Wasson et al. 2017) that are vulnerable to degradation by invasive species (Bertness et al. 2002), shifts in community composition (Boyer & Zedler 1999), and high levels of macroalgal production that reduce water quality (Jeppesen et al. 2018) and cause channel bank instability (Wasson et al. 2017). Due to these stressors, wetland resilience to global change (i.e., sea-level rise; Kennish 2002; Kirwan & Megonigal 2013, drought; Silliman et al. 2005) is compromised. Globally, over 60% of coastal wetlands have been lost (Davidson 2014). Along the U.S. West Coast estimates of coastal wetland loss are as high as 85% (Brophy et al. 2019), with model projections for high end of

Author contributions: SS, HMP conceived and designed the studies; KMB wrote the manuscript with input from all authors; KMB, SS analyzed the data; SS, MP, AD supervised the work; AD collected and processed the data; KMB, SS, DR, RS, MP aided in interpreting the results; KMB, SS contributed equally to this work and are co-first authors.

¹Marine Science Institute, University of California, Santa Barbara 93106, U.S.A.

²Address correspondence to K. M. Beheshti, email katbeheshti@ucsb.edu

© 2023 The Authors. Restoration Ecology published by Wiley Periodicals LLC on behalf of Society for Ecological Restoration.
This is an open access article under the terms of the [Creative Commons Attribution License](#), which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

doi: 10.1111/rec.13936

Supporting information at:

<http://onlinelibrary.wiley.com/doi/10.1111/rec.13936/supinfo>

the century sea-level rise predicting future losses of 68–100% (Thorne et al. 2018).

Concerns about the accelerating loss of coastal wetlands have increased efforts to protect and restore, and create wetland habitat to reverse declines and compensate for historical reclamation (Dobson et al. 1997; Zedler & Kercher 2005). Yet, restored and newly created wetlands can take decades to resemble and function similarly to natural wetlands (Zedler & Kercher 2005; Moreno-Mateos et al. 2012; Sullivan et al. 2017), largely because wetland vegetation, which provides the structure necessary to support key ecological functions and processes (e.g. storm buffering, carbon sequestration, and habitat provisioning), is often slow to establish (Callaway 2005; O'Brien & Zedler 2006; Moreno-Mateos et al. 2012). Following restoration, the natural establishment of marsh plant species of similar composition and cover to reference habitats can take decades (Wolters et al. 2008; Jin et al. 2021) to a century (Byers & Chmura 2007).

In mature salt marshes, tidal hydrology, sediment characteristics, and biological interactions largely determine the cover and species composition of salt marsh vegetation. Typically, the lower limits of vegetation are set by tidal inundation frequency (waterlogging), whereas the upper limits are set by intolerance to soil salinity (Zedler & Beare 1986) or species interactions (competition; Clements 1916; Bertness & Ellison 1987, facilitation; He & Cui 2015). Throughout established salt marshes, physical stressors and species interactions determine plant zonation patterns (Pennings & Callaway 1992; Pennings et al. 2005; Cui et al. 2011). In particular, in semiarid estuaries, soil salinity and freshwater pulses control plant establishment and distribution (Zedler et al. 1986). However, less is known about the role of these stressors (hypersalinity, poor drainage, and drought) in affecting marsh plant establishment on newly restored or created habitat with little to no existing vegetation, especially in semiarid environments (but see Zedler et al. 2003; Zedler & Kercher 2005; Thomsen et al. 2022).

On a newly restored, bare landscape, physical stressors (e.g. desiccation and high soil salinity) may limit natural recruitment of salt marsh plants and negatively affect their survival by restricting the “window of opportunity” for seedling establishment (Balke et al. 2014; Hu et al. 2015). As autogenic ecosystem engineers (Jones et al. 1994), salt marsh plants affect key ecological processes (sediment dynamics and hydrology; Bouma et al. 2005, 2010) and introduce habitat complexity that supports important functions (biodiversity; Whitcraft & Levin 2007, nutrient cycling; Sousa et al. 2010) via their own physical structure. Therefore, poor establishment of salt marsh plants following restoration may delay or disrupt desired functions. Active interventions to improve abiotic conditions may help facilitate vegetation development. Such interventions include irrigating with freshwater to reduce soil salinities and increase soil moisture, removing barriers in impounded areas or regrading to reestablish tidal flow and improve drainage, and tilling and amending compacted and nutrient-poor soils (O'Brien & Zedler 2006; Brooks et al. 2015; Nolan et al. 2020). Actively seeding (Yuan et al. 2020) and planting salt marsh plants (Zedler et al. 2003) can also enhance vegetation development.

However, the efficacy of these methods can vary within and among systems (Zedler & Callaway 1999; Zedler & Kercher 2005). Therefore, restoration programs will likely benefit from in-situ studies that examine the factors that hinder and facilitate the development of salt marsh vegetation because these studies help minimize uncertainty in methods and increase the likelihood of attaining the preestablished restoration goals (Zedler 2017). Iterative, hypothesis-driven field experiments that identify the most effective management pathway for a particular restoration site are referred to as “adaptive restoration,” which is an important approach for advancing restoration science (Zedler 2017). Adaptive restoration allows practitioners to identify site-specific stressors and then test which management strategies are most effective at ameliorating primary stressors.

Many wetland restorations involve grading upland or reclaimed soils that are physically or chemically different from natural salt marshes (Thomsen et al. 2022). Grading typically compacts soils and creates a smooth landscape that lacks the microtopography and habitat heterogeneity characteristic of natural wetlands (Thomsen et al. 2022). A smooth, compacted marsh plain with too gradual of a slope is associated with poor drainage, ponding, and poorly aerated, hypersaline soils that inhibit the establishment and growth of marsh plants (Zedler et al. 2003; Brooks et al. 2015). In natural marshes, accretion and erosion create habitat heterogeneity. Yet, in restored high marshes, compact sediments are less likely to erode and rates of sedimentation are low. Experimentally augmenting habitat heterogeneity can increase drainage and aeration of marsh soils, allowing seedlings to establish and existing marsh plants to grow (Xie et al. 2019b; Fivash et al. 2020). Active restorations that use potted plants from nurseries offer both localized decompaction and soil amendments.

Amending restored salt marsh soils to ameliorate stressful conditions is another approach with variable results within and among sites. Some studies have shown no sustained growth benefits of soil amendments (Vance et al. 2003; Watson et al. 2015), while others have shown greater marsh plant biomass, growth, and branching with amendments (Boyer et al. 2001; O'Brien & Zedler 2006; Foster-Martinez & Variano 2018). Depending on the amendments applied, there can also be trade-offs of reducing soil salinities and enhancing cover of both native and non-native species (Nolan et al. 2020). Given these variable results, site-specific studies that evaluate the efficacy of different restoration methods previously tested elsewhere (e.g., amending, decompacting, planting, seeding, irrigating) are a prudent prerequisite for establishing marsh vegetation at a restored site.

In this study, we experimentally evaluated potential factors causing the slow development of salt marsh vegetation at the San Dieguito Wetlands Restoration Project (hereafter referred to as San Dieguito Wetlands), a semiarid wetland restoration in Del Mar, California (Fig. 1). Over a four-year period (2019–2022), we examined the following questions to determine the stressors inhibiting the development of marsh vegetation post-restoration and inform best practices for facilitating plant establishment and growth:

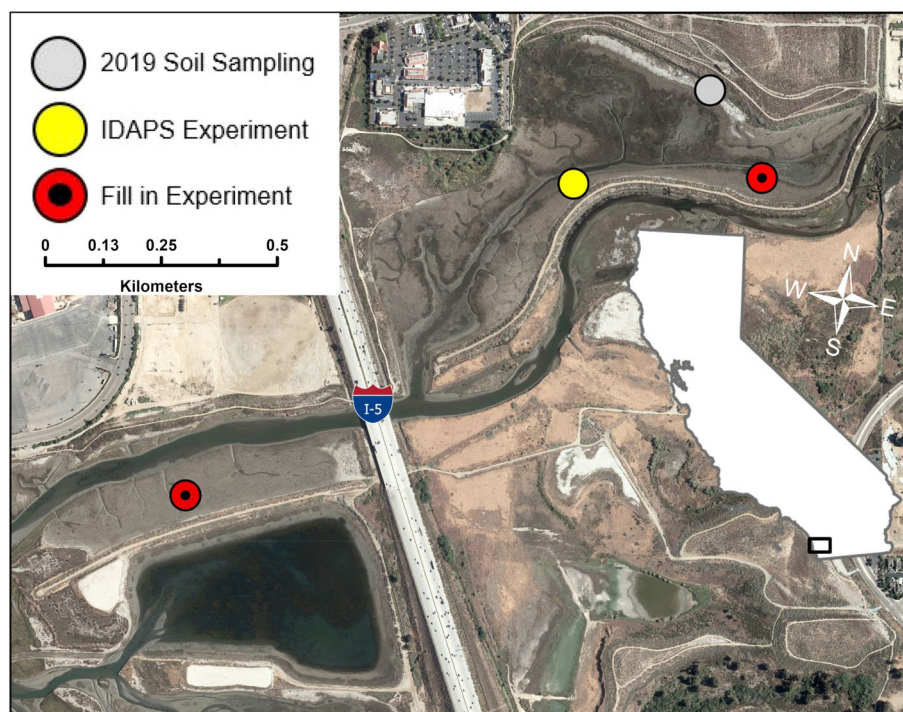


Figure 1. Study site map. In 2019, soil samples were collected from an area (gray circle) with similar characteristics to the IDAPS and Fill In locations. The IDAPS (yellow circle) was conducted northeast of I-5 and the Fill In (red circles) was conducted east of the IDAPS and west of the I-5. Both studies fell within the San Dieguito Wetland Restoration Project area. Source: Ocean Imaging.

- (1) What are the primary stressors inhibiting the establishment of marsh plants in the mid to high marsh?
Does existing vegetation lessen the severity of these stressors?
- (2) What forms and/or combinations of intervention (irrigation, decompaction, amendments, planting, and seeding) are most effective at establishing marsh plants and increasing cover in the high marsh?
- (3) Is planting or seeding more effective in the mid-marsh, and is relative plant performance species-dependent?

Collectively, these investigations helped inform the adaptive restoration and management of San Dieguito Wetlands.

Methods

Study Site

San Dieguito Lagoon (32.969545°N, 117.26047°W) is an urban estuary in Del Mar, California, U.S.A. (Fig. 1), 32 km north of San Diego (Fig. 1). The Mediterranean climate is semiarid and rainfall is seasonal, averaging 28 cm/year (Elwany 2011). Tides in the estuary are semidiurnal with a spring tidal range of 1.2 m and a neap tidal range of 0.5 m. The San Dieguito River flows through the estuary and eventually empties into the Pacific Ocean at its inlet, which is dredged regularly to maintain tidal flushing.

Prior to its restoration, this highly degraded tidal wetland was estimated to have lost over 70% (256 acres) of its

historical habitat (Kent & Mast 2005). Restoration efforts beginning in 2008 have attempted to restore 150-acres, including 93 acres of salt marsh. A decade later, vegetation cover remained patchy in all but a few areas. Over this period, the wetland averaged a 2.8% increase in cover annually (Fig. 2). Despite numerous plantings and installation of freshwater irrigation over large areas of the wetland, vegetation cover has increased by only 30% wetland-wide since 2011 and remains ~23% below the target set by reference wetlands (Fig. 2). At this rate, without further intervention, it would take an additional eight or more years for the wetland to reach the vegetation cover targets set by reference sites within the region (Fig. 2; Supplement S1).

Characterizing Soil Conditions

We collected soil samples at San Dieguito Wetlands in areas with persistently low cover of salt marsh vegetation (Fig. 1) to identify potential stressors to vegetation that could be manipulated and tested in subsequent field experiments. We collected soil samples at four locations spaced 10–20 m apart at four elevations (1.61, 1.76, 1.92, and 2.07 m MLLW) in August 2019. At each sampling location ($n = 16$), four soil cores (3.5 cm diameter $\times$ 10 cm deep) were collected and combined from vegetated and nonvegetated patches ($n = 64$ cores). We sent the samples to the University of California, Davis Soil Laboratory for bulk density, soil salinity and soil moisture content analysis. Bulk density was calculated as the dried (105°C) mass of a

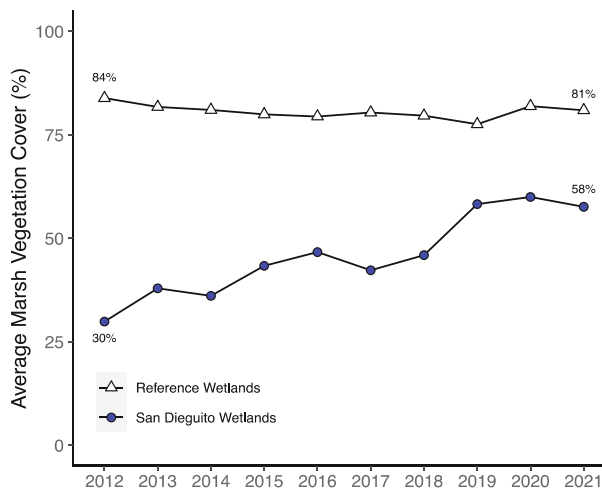


Figure 2. Marsh vegetation cover at San Dieguito and reference wetlands (2012–2021). Marsh vegetation cover (%) at San Dieguito (blue circles) increased steadily over the past 10 years, but has yet to meet targets set by reference sites (white triangles). The annualized rate of increase in vegetation cover is 2.8% per year at San Dieguito. At this rate, it would take an additional 8+ years for the restoration site to reach the cover of reference wetlands.

known volume of soil (g/cm^3). For soil salinity, samples were first saturated with water and the concentration of dissolved salts (ppt) were measured from paste-extracted water. For soil moisture content, samples were saturated and allowed to equilibrate under a preset pressure potential to derive percent (%) water retention via moisture release curves (Klute 1986).

We used general linear models to test the main effects of: (1) tidal elevation and patch type (vegetated vs. nonvegetated) on soil salinity and soil moisture content and (2) bulk density and patch type on soil moisture content. If patch type was significant, we assessed differences between vegetated and nonvegetated patches in soil salinity, soil moisture, or bulk density, using a two-sample *t*-test. The results from the soil analyses informed the choice of treatments for the experiments by identifying potential stressors (high soil salinity and compaction, low soil moisture) that could inhibit marsh vegetation establishment and growth at mid-to-high tidal elevations. To further contextualize the results from the soil sampling, we used data collected from HOBO Water Level Loggers to derive percent time submerged relative to tidal elevation (Fig. S1A). All analyses, unless otherwise stated, were conducted in R version 4.2 (R Development Core Team 2022).

IDAPS Experiment

The IDAPS experiment started on March 3, 2020 in the north-eastern portion of the restoration site (Fig. 1). Vegetation cover within the study area was 5% or less and fell within 1.61–2.07 m MLLW (Figs. 1 & S1). There were three replicate blocks, each of which included an irrigated (I+) and not irrigated (I−) section and three soil treatments: (1) decompacted and amended (D+A+), (2) decompacted only (D+A−), and

(3) neither decompacted nor amended (D−A−; Fig. 4). Soil amendments were tilled into the soil and consisted of a mix of gypsum and mulch (Kellogg Gromulch 2-in-1 Planting Mix & Mulch Organic Plus). The mulch was enriched with fertilizer and included recycled forest material, manure, and feather meal. In addition to the irrigation and soil treatments, each block had the following treatments applied to the 1.5×3.0 m experimental plots: (1) planting and no seeding (P+S−) of nursery-grown glasswort (*Arthrocnemum subterminale*) where 10 1-gallon *A. subterminale* plants were spaced 0.6 m on center in two rows of five, (2) seeding and no planting (P−S+) with a mix of seeds from four species (*Salicornia pacifica*, *Frankenia salina*, *A. subterminale*, *Distichlis spicata*), and (3) control plots with neither planting or seeding (P−S−). Each section had a control plot, D−A−P−S− in irrigated (I+) and not irrigated (I−) sections (Table S1; Fig. 4A). After detecting no net positive effect of seeding (Fig. S2; $\chi^2 = 0.639$, $p = 0.42$), we summarized and analyzed data without consideration of seeded (S+) treatments, which failed to produce any seedlings (but see Fig. S2). All reported results exclude seeded plots and include all of the 12 remaining treatment combinations (i.e. irrigation, decompaction, amendment, and planting).

We conducted surveys of percent cover and plant area quarterly from March 2020 to January 2022. We collected percent cover data using a uniform point contact (UPC) method. At each $1.5 \text{ m} \times 3.0 \text{ m}$ plot, a rope with 14 knots was extended across the 3.0 m width of the plot at 0.2 m intervals (Fig. S3a). The species present under each of the 98 points in a plot were recorded and used to estimate the combined percent cover of all species present within each plot for each survey. Data collected at the species level were used to identify dominant and rare species within each plot. We collected additional data in planted plots (P+) beginning in October 2020 to track the survivorship and size of planted individuals through time. During each survey, we identified individuals as either alive or dead based on the presence of living aboveground biomass. Individual size was measured as the radius of the maximum length (“a”) and width (“b”) of the plant, estimates of plant area (m^2) were calculated assuming an elliptical shape, using the equation $A = \pi \cdot a \cdot b$ (Fig. S3b).

To evaluate whether treatment effects on total vegetation cover varied with time, we ran a full factorial repeated measures mixed model (MMRM) with an unequal variance covariance structure, using the experimental plots as the repeated structure. The experimental plots were used as the repeated structure to account for any potential carry-over effects between surveys and track plot-level treatment responses over time. The fixed effects included time since the start of the experiment, irrigation, soil treatment, and planting, with all possible interactions included in the model and the random effect of block. To evaluate differences ($\alpha < 0.05$) between model effects, we ran Tukey’s HSD post-hoc test for each model effect of interest. We ran all analyses associated with the MMRM using JMP Pro 16 (JMP® 1989–2021).

To determine whether there was an effect of treatment on final vegetative cover at the end of the experiment, we ran a mixed model with a restricted maximum likelihood (REML)

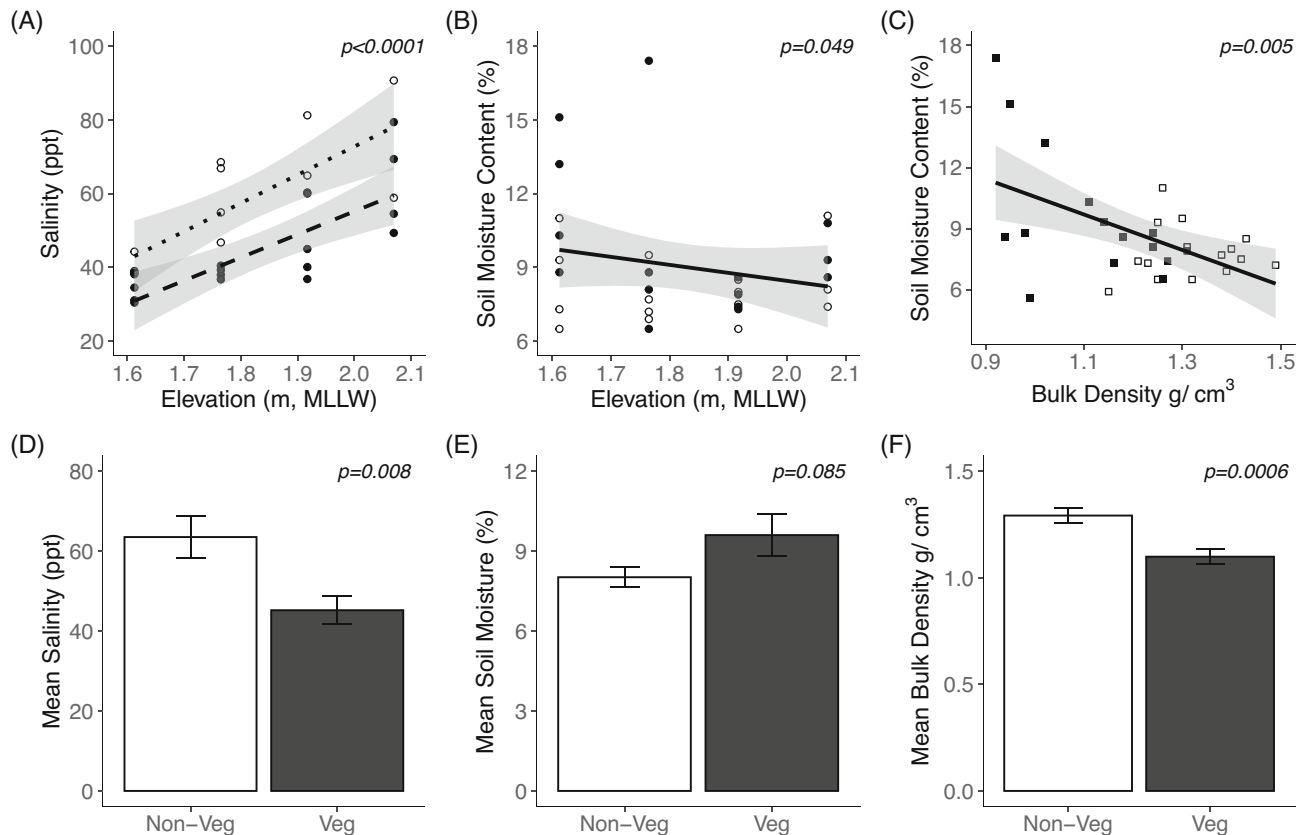


Figure 3. Mid to high marsh soil properties at San Dieguito Wetlands. (A) Salinity increased and was higher across elevations in nonvegetated (unfilled circles, dotted line) patches relative to vegetated (filled circles and dashed line) patches. Soil moisture content (%) decreased with (B) elevation and (C) bulk density. Figure (A) indicates that the relationship between elevation and salinity differed between patch types (nonvegetated vs. vegetated). Data plotted as a function of elevation (A,B) is shown as circles and data plotted as a function of bulk density (C) as squares, with 95% CI. Filled points and bars represent soil samples collected in vegetated ("Veg") patches and unfilled points and bars represent soil samples collected in nonvegetated ("Non-Veg") patches (D–F). Data are plotted as the mean ($\pm$ SE) salinity (D), soil moisture content (E), and bulk density (F). Reported in the upper left-hand corner are p-values for each mixed model (A–C) and two-sample *t*-test (D–F). Data were collected in 2019.

estimation method in R version 4.2 (R Development Core Team 2022). We summarized the survey data collected in January 2022 (the last survey) at the quadrat scale by summing cover of all species per quadrat. The full factorial model included the fixed effects of irrigation, soil treatment, and planting, and the random effect of block. To evaluate differences between model effects ($\alpha < 0.05$), we ran Tukey's HSD post-hoc test.

During our field surveys, we noticed high cover of *S. pacifica*, which was not planted. To assess how various treatment combinations affected species-specific cover, we analyzed the final survey data (January 2022) using a generalized regression with a beta distribution and maximum likelihood estimation method and logit model link using the *betareg* package in R version 4.2 (R Development Core Team 2022; Cribari-Neto & Zeileis 2010). For the generalized beta regression, we used the main model effect of IDAP (irrigation, soil treatment, planting) and the response variable of species-specific cover, transformed to meet the assumptions of a beta distribution (range of 0–1).

Fill In Experiment

Concurrent with the IDAPS experiment, the Fill In experiment focused solely on planting versus seeding of three species—*S. pacifica*, *F. salina*, and *A. subterminale* at two locations in the mid marsh (1.6–1.7 m MLLW; Fig. 1). Similar to the IDAPS, each plot in the Fill In experiment measured 1.5×3.0 m. Planted treatments included 10 1-gallon plants of a single species (*S. pacifica*, *F. salina*, or *A. subterminale*), placed in the ground in two rows of five (Fig. 4B). There were six replicate plots per treatment, including seeded plots (seed mix including: *S. pacifica*, *F. salina*, *D. spicata*, and *A. subterminale*) and unmanipulated controls (Table S2). We measured the cover of marsh vegetation and planting area quarterly from March 2020 to January 2022 following the same methods as the IDAPS described above.

To evaluate whether treatment effects on vegetation cover varied over time, we ran a full factorial MMRM with an unequal variance covariance structure, using the experimental plots as the repeated structure. The fixed effects included time since the start of the experiment, planting treatment, and their

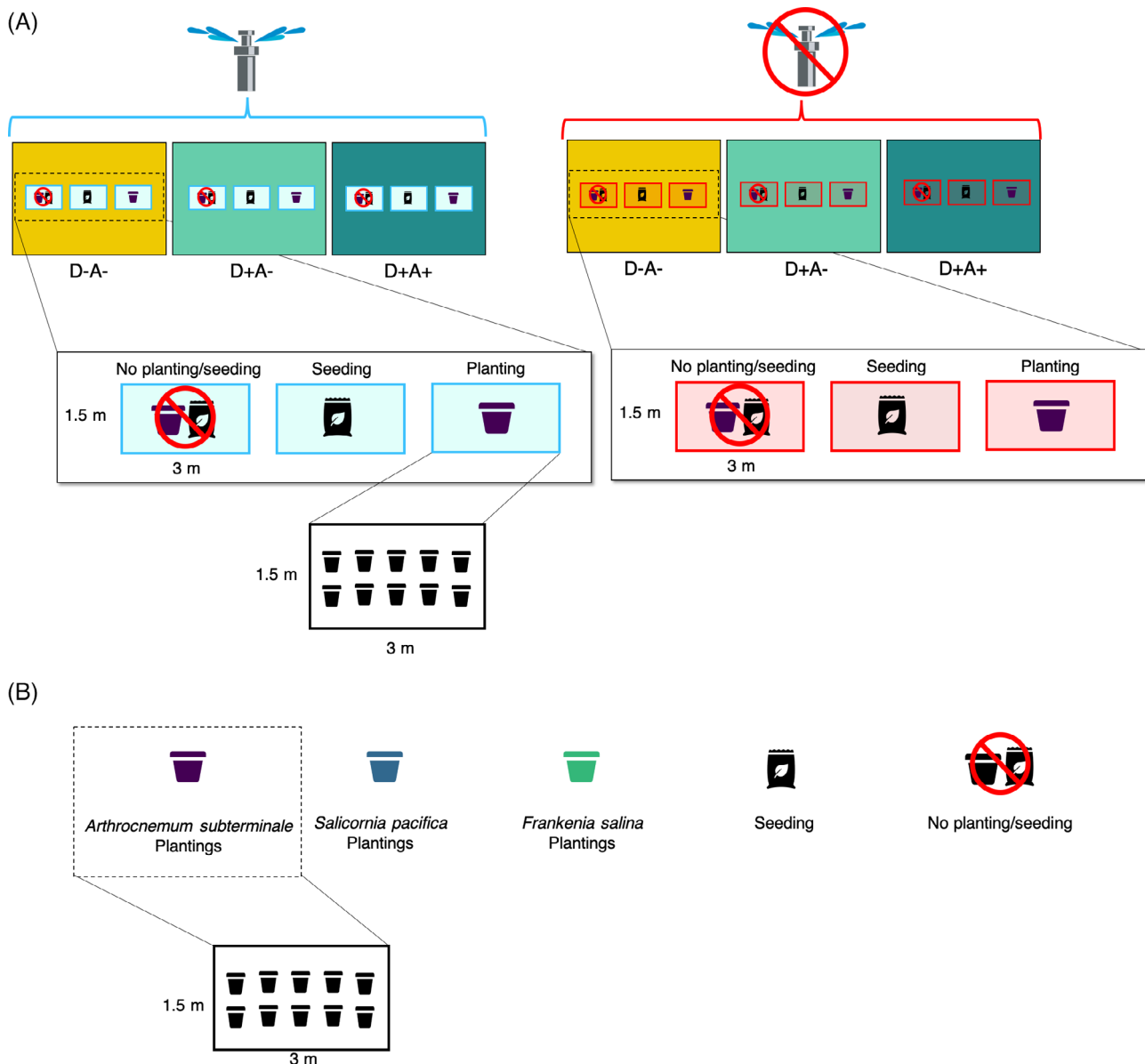


Figure 4. Schematic of the (A) IDAPS and (B) Fill In experiments at the block level. (A) The IDAPS (“I” = irrigation, “D” = decompaction, “A” = amendment, “P” = planting, “S” = seeding) had three blocks and each block had an irrigated and unirrigated section. Within each section were three sub-blocks with (D+A– or D+A+) or without (D–A–) soil treatments. In each sub-block were three 1.5 × 3.0 m plots with one of three treatments (1) no planting or seeding, (2) seeding, or (3) planting. For planted treatments, 10 1-gallon plots of *Arthrocnemum subterminale* were planted in two rows of five, spaced 0.6 m apart on center. (B) For the Fill In, there were four treatments and one control (no seeding or planting) per block. Within each of the six blocks were the five plot types (*Arthrocnemum subterminale* planting, *Salicornia pacifica* planting, *Frankenia salina* planting, seeding, and no planting/seeding). Plantings in the Fill In were installed in two rows of five, spaced 0.6 m apart on center.

interaction and the random effect of location. To evaluate differences ($\alpha < 0.05$) between model effects, we ran Tukey’s HSD post-hoc test for each model effect of interest. Similar to the IDAPS analysis, we ran the MMRM and associated post-hoc tests using JMP Pro 16 (JMP® 1989–2021).

To determine whether there was an effect of treatment on final vegetative cover, we ran a mixed model with a REML estimation method in R version 4.2 (R Development Core Team 2022),

similar in structure to the analysis performed on the IDAPS data. We summarized data collected from the final survey in January 2022 at the quadrat scale by summing cover of all species per quadrat. The model included the fixed effect of treatment (*Arthrocnemum* Planting, *Salicornia* Planting, *Frankenia* Planting, Seeding, and Control) and the random effect of location. To evaluate differences ($\alpha = 0.05$) between treatments, we ran Tukey’s HSD post-hoc test. During our field surveys, we noted

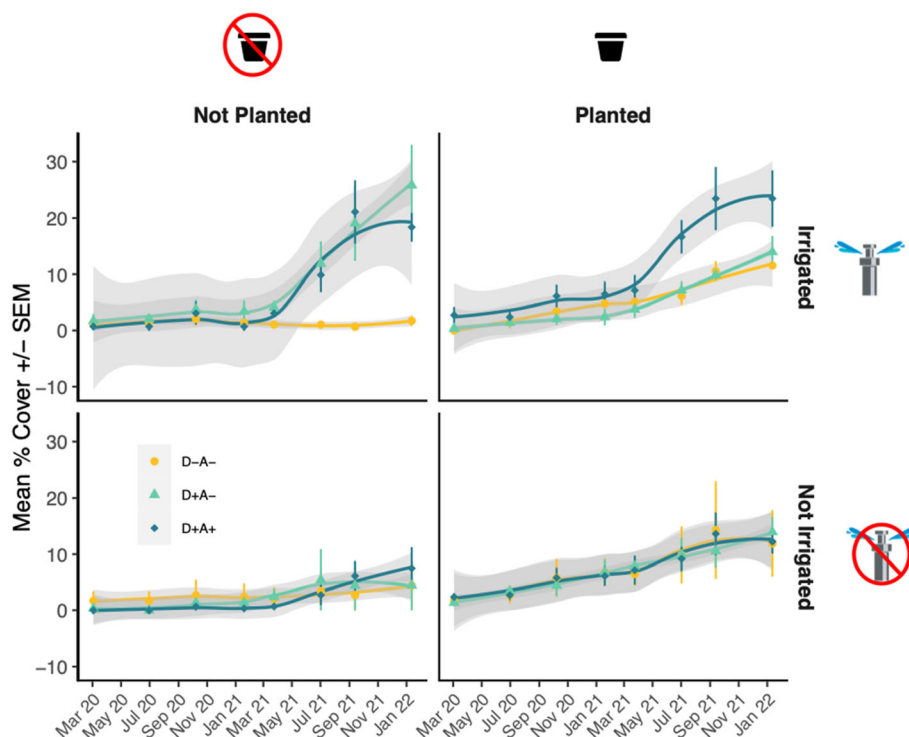


Figure 5. Plotted is the mean ($\pm$ SE) vegetation cover by treatment from March 2020 to January 2022 fit with a loess regression (95% CI). Treatments included soil treatments with (+) or without (–) decompaction (“D”) and soil amendments (“A”) and were either not planted (left) or planted (right) with 10 1-gallon plants of *Arthrocnemum subterminale* and either irrigated (top) or not irrigated (bottom). Soil treatments are represented as follows; D–A– as gold circles, D+A– as turquoise triangles, and D+A+ as dark blue diamonds.

that the cover of planted plots was dominated by the species planted, rather than species recruiting into the plots. For this reason, we did not analytically explore how treatments affected species-specific cover as was done for the IDAPS.

Results

Soil Properties in Mid-to-High Marsh

Soil conditions were more stressful in the high marsh than in the mid marsh and in nonvegetated patches than vegetated patches. Higher tidal elevations had higher salinities (>60 ppt) and lower soil moisture (Fig. 3A & 3B), the latter of which was negatively correlated with bulk density (Fig. 3C). Nonvegetated patches had higher soil salinities (Fig. 3D), lower soil moisture content (Fig. 3E) and higher bulk density (Fig. 3F) than vegetated patches. Soil salinity increased significantly with elevation ($R^2 = 0.72$, $F_{[2,29]} = 36.81$, $p < 0.0001$) and the slope of the relationship was greater in nonvegetated (slope = 30.6) patches compared to vegetated patches (slope = 19.6) patches (Fig. 3A). Soil moisture content had a stronger relationship with bulk density (Fig. 3C; $R^2 = 0.31$, $F_{[2,29]} = 6.39$, $p = 0.005$, slope = –8.3) than elevation (Fig. 3B; $R^2 = 0.19$, $F_{[2,29]} = 3.36$, $p = 0.049$, slope = –1.4), irrespective of patch type.

Salinity, on average, was 40% higher in nonvegetated (63 ppt) patches compared to vegetated (45 ppt) patches

(Fig. 3D; $t(25.9) = 2.89$, $p = 0.008$). Soil moisture in nonvegetated patches did not differ significantly from vegetated patches (Fig. 3E), while bulk density averaged 18% higher in nonvegetated patches compared to vegetated patches (Fig. 3F; $t(29.9) = 3.82$, $p = 0.0006$). Collectively, these results suggest that soil conditions were likely more stressful (hypersaline and less saturated) at higher elevations than at lower elevations, particularly in nonvegetated areas.

Patterns of Vegetative Development Facilitated by IDAPS Treatments

Average vegetation cover in the IDAPS experiment increased 10-fold over the 2-year study (Table S3; Fig. 5). Differences in cover between treatments emerged in the second year of the experiment (March 2021–January 2022). Irrigation and soil treatments (i.e. decompaction and amendments) increased vegetation cover, and the importance of both factors varied over time (Table S3), strengthening in year two.

After 2 years, the vegetation cover in individual plots of the IDAPS experiment reached as high as 37%. Vegetation cover was significantly higher in not planted plots that were irrigated and decompacted (I+D+A–P–), compared to plots that were irrigated, but neither decompacted nor amended (I+D–A–P–) and plots that were not irrigated and amended (I–D–A–P–, I–D+A–P–) (Fig. 6A). There were no differences in final vegetation cover among treatments in planted plots (Fig. 4A).

Vegetation cover in planted (14%) plots was slightly, but not significantly ($p = 0.088$) higher than not planted (10%) plots (Table S4; Fig. 6). Final vegetation cover was also strongly influenced by irrigation and soil treatments (Table S4). A post-hoc Tukey test showed that cover in irrigated plots was greater than not irrigated plots—cover in irrigated plots was 54%

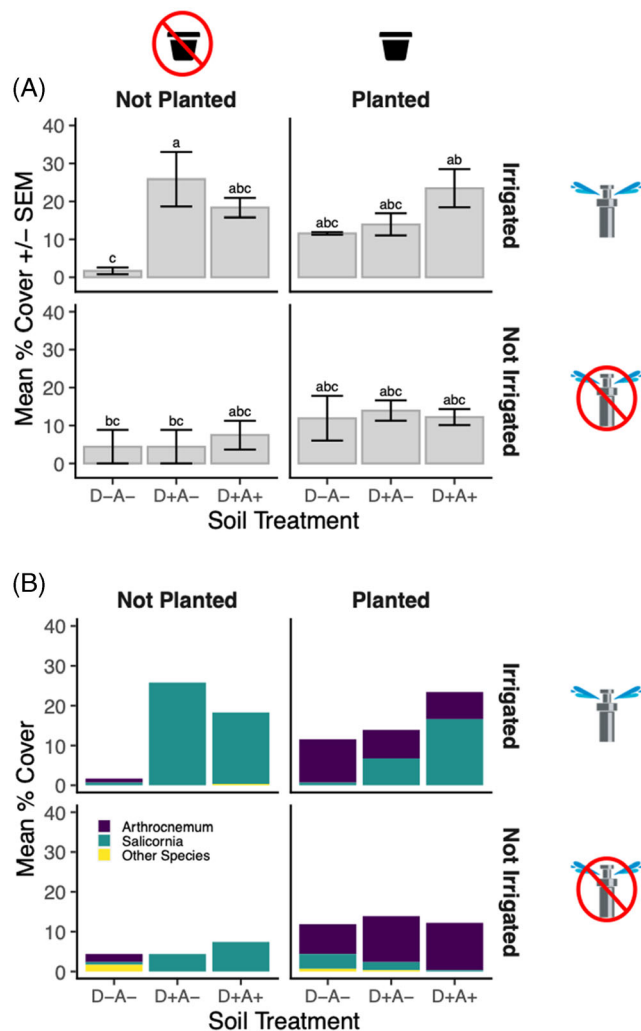


Figure 6. Plotted is the mean vegetation percent cover (A) across all species (gray bars) and (B) by species, at the last survey date in January 2022 (~22 months after the study began) for each soil treatment (with (+) or without (–) decompaction (“D”) and soil amendments (“A”)) that was not planted (left) or planted (right) with 10 1-gallon plants of *Arthrocnemum subterminale* and either irrigated (top) or not irrigated (bottom). Gray bars (A) represent mean ($\pm$ SE) vegetation cover when averaged across all species. Mean vegetation cover (A) was highest and comparable between planted treatments and not planted, irrigated treatments with soil treatments (D+A–, D+A+) applied. Different letters denote significant ($\alpha < 0.05$) differences between treatment combinations. The relative contributions to mean cover by species (B) are color coded with *Arthrocnemum* in purple, *Salicornia* in teal, and Other Species (i.e. *Distichlis spicata*, *Frankenia salina*) in yellow. Irrigation, in combination with soil treatments, facilitated the natural recruitment and growth of *Salicornia*, while *Arthrocnemum* was close to absent from plots where it was not planted (B).

greater than not irrigated plots. Additionally, cover in decompacted and/or amended (D+A+, D+A–) plots was 65–70% greater than plots that were not decompacted or amended (D–A–; Table S4; Fig. 6). Excluding planted plots, average vegetation cover in not seeded plots (P–S–; 10%) was over two times the cover of seeded plots (P–S+; 4%; Fig. S2).

The species composition of marsh plants and their relative contributions to total vegetation cover varied by treatment. *A. subterminale* dominated overall vegetation cover when planted, with one exception (irrigated, decompacted, amended, and planted plots) where average cover of the marsh dominant *S. pacifica* was 71% (Fig. 6B). Naturally recruited *S. pacifica* dominated vegetation cover in not planted plots. Irrigation with decompaction facilitated *S. pacifica* cover (Fig. 4B, S4a; $\chi^2 = 16.8$, $p = 0.003$, $R^2 = 0.47$), but these treatments did not affect cover of planted *A. subterminale* (Fig. 4, S4b; $\chi^2 = 137.1$, $p < 0.0001$, $R^2 = 0.71$).

Variation in Vegetation Cover Explained by Species Planted in the Fill In Experiment

Average vegetation cover increased by 13–22% across all planted species whereas cover increased by only 2–3% in seeded and control plots (Table S5; Fig. 7). While we standardized pot size (1-gallon) of planted individuals, the cover of planted plots at the beginning of the experiment was frequently higher in the *Frankenia* (4–13%) and *Salicornia* (3–18%) plots than *Arthrocnemum* (0–8%) (Figs. 7 & S5). Vegetation cover at the beginning of the experiment was 0–7% for seeded plots and 0–5% for control plots (Fig. 7). Despite differences in the initial cover, the relative change in cover varied across the treatments with a 2%, 3%, 12%, 19%, and 22% increase in cover in Control, Seeding, *Arthrocnemum*, *Salicornia*, and *Frankenia* Plantings,

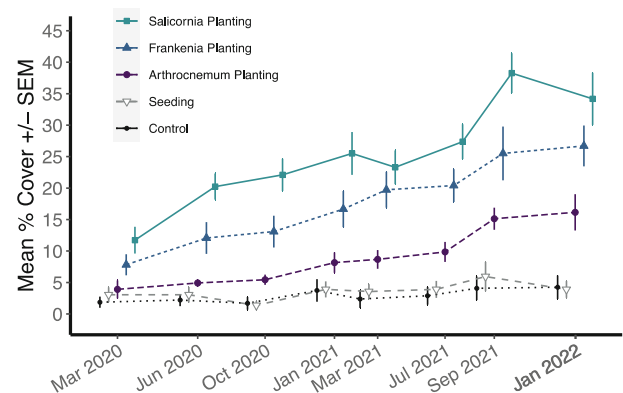


Figure 7. Mean ($\pm$ SE) vegetation cover increased steadily over time for all planted treatments relative to the seeded or control treatments. Points are dodged to reduce overlap of points and error bars. Treatments are displayed as follows: *Salicornia* = teal squares, solid line; *Frankenia* = blue triangles, short dashed line; *Arthrocnemum* = purple circles, dashed line; seeding = white down-facing triangles, long dashed line; control = black circles, dotted line.

respectively (Fig. 7). The effects of treatment on vegetation cover varied over time (Table S5). Cover was consistently low in seeded treatments and controls, but gradually increased over time in all the planted treatments with *Salicornia* plots attaining the highest cover, *Frankenia* intermediate and *Arthrocnemum* the lowest of the three species (Figs. 7 & S6).

At the end of the 2-year Fill In experiment the maximum cover of vegetation for an individual plot was 47% with cover varying significantly among the treatments (Fig. 8A; $F_{[4,24]} = 22.18$, $p < 0.0001$). For example, average vegetation cover in the *Salicornia* Planting treatment was more than twice that of the *Arthrocnemum* Planting treatment ($p < 0.05$, post-hoc Tukey test). Vegetation cover in seeded and control plots was the same at 4% (Fig. S2), while cover in planted plots was over six times as great at ~26% (March 2020–January 2022). All planting treatments had significantly higher cover than the seeding treatment and controls; there were no differences in cover between *Salicornia* and *Frankenia* Planting treatments or *Frankenia* and *Arthrocnemum* Planting treatments (Tables S5 & S6; Fig. 8A).

The identity of the species planted largely determined plot-level species composition (e.g. *A. subterminale* made up the majority of cover in plots planted with *Arthrocnemum*). *S. pacifica* was the dominant species in the seeded and control plots and was generally the only other species that naturally recruited into treatments where it was not planted (Fig. 8B). In plots planted with *Salicornia* it made up 100% of the observed cover (Fig. 8B). Planting survival was high for the duration of both the IDAPS (>87%) and Fill In (>93%) experiments and we detected differences in the rate of planting growth and expansion between and within species in both experiments (Fig. S5). See Supplement S1 for additional results on plant survival and growth.

Discussion

Stressful Environmental Conditions Affect Rates at Which Restoration Targets are Met

In salt marsh restoration, plant establishment is often slowest in the mid to high marsh, the target area of our investigations, but the drivers of this slow vegetation growth are often unclear and vary with local conditions. In a semiarid wetland, we found that salinity stress was greatest in unvegetated, high marsh areas, which suggests that plants either enhance conditions that favor their survival and growth or establish in more hospitable patches throughout the mid to high marsh. These results are consistent with previous findings that engineering effects of existing vegetation (i.e. shading, He & Cui 2015; Whitcraft & Levin 2007; macropores, van Putte et al. 2020) can ameliorate the stress of high salinity, low soil moisture, and compact soils, and facilitate the establishment of other marsh plants. Leveraging positive species interactions to facilitate restoration success has proven fruitful in certain contexts (Byers et al. 2006; Silliman et al. 2015; Fischman et al. 2019), especially in stressful environments (Bertness & Callaway 1994). However, restoration outcomes are variable (Fischman et al. 2019; Tanner et al. 2022), demonstrating the need for site-specific studies to inform restoration design. Building on these findings, the IDAPS and Fill In experiments tested the effects of both the direct amelioration of stressors (i.e. high soil salinity and compacted soils) and the indirect benefits of vegetation (i.e. planting and seeding) on establishing and hastening growth and cover of marsh vegetation that was pre-existing, planted, or newly recruited. Our results highlight the limits of marsh plants as ecosystem engineers (Jones et al. 1994) and positive species interactions in encouraging plant establishment (i.e. planting only) and the potential need for more intrusive methods of intervention

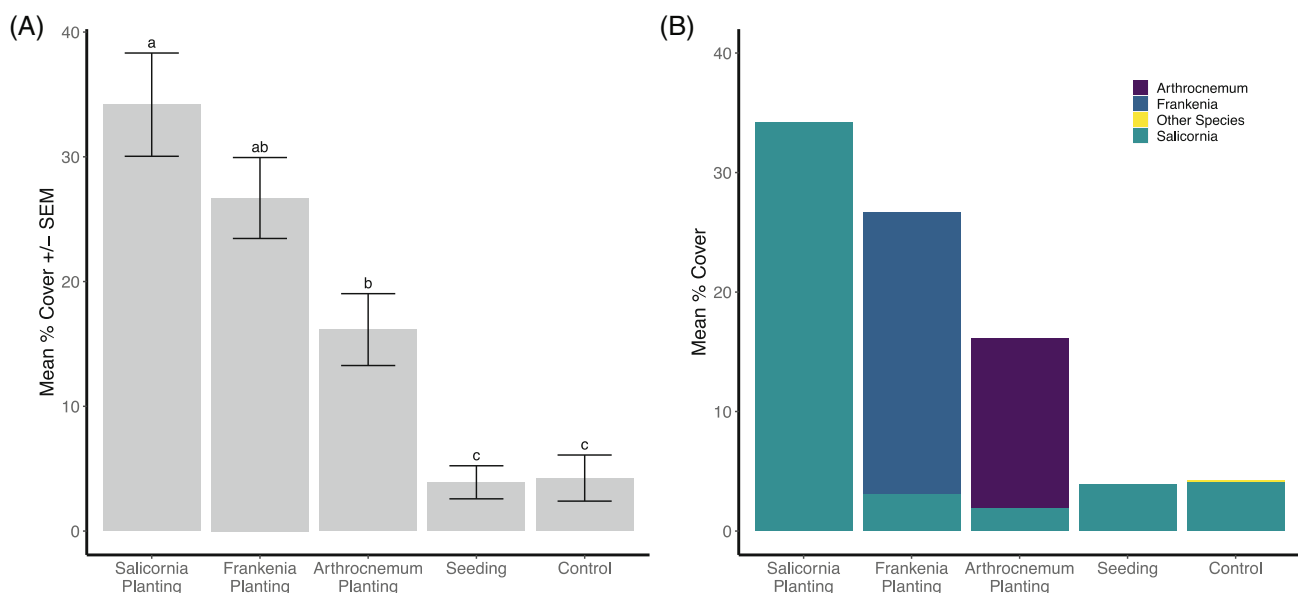


Figure 8. (A) Mean ($\pm$ SE) vegetation cover by treatment with different letters denoting significant ($\alpha < 0.05$) differences between treatments and (B) mean vegetation cover with the contributions to overall mean cover color coded by species. In January 2022, species other than *A. subterminale* (purple), *F. salina* (blue), and *S. pacifica* (teal) made up ~1% or less of the cover for any given treatment (yellow, "Other Species").

(i.e. large-scale irrigation and decompaction). Without such intervention, it may take another decade for San Dieguito Wetlands to reach the vegetation cover targets set by reference wetlands. Many wetland restoration projects have followed similarly slow trajectories to what we observe at San Dieguito Wetlands (Zedler & Callaway 1999; Byers & Chmura 2007; Thomsen et al. 2022). Therefore, restoration planning and design should consider the initial conditions necessary for plant establishment prior to implementation to ensure vegetation cover targets are met.

Effects of Planting, Irrigation, Decompaction, Soil Amendments, and Seeding in the Mid and High Marsh

Planting can be an effective strategy for increasing vegetation cover, but the ability of plants to establish and spread depends on the species growth form. Therefore, plant identity should be considered in restoration planting programs, particularly if a restoration goal is to increase cover rapidly. Our results complement previous studies by demonstrating that planting fast-growing marsh species (e.g. *Salicornia pacifica*) is most effective at increasing lateral spread and introducing structure to an otherwise bare landscape (Yando et al. 2019; Howard et al. 2020). Fast-growing species have also been shown to increase inter- and intra-specific plant survival rates (Silliman et al. 2015; Howard et al. 2020) and increase species richness in stressful environments (Xie et al. 2019a, 2019b). Although some managers may caution against planting fast-growing species like *S. pacifica* that can quickly dominate a marsh landscape, the short-term disadvantage of lower plant diversity (Callaway et al. 2003) may be countered by improved soil conditions from the establishment of these faster growing ecosystem engineers (Jones et al. 1994). Attaining vegetation cover comparable to reference wetlands is an explicit restoration target at San Dieguito Wetlands, which is a major challenge in the mid to high marsh. Our results illustrate the potential short-term tradeoff between cover and diversity, given our explicit restoration target to achieve comparable vegetation cover to reference wetlands. Longer-term field experiments are needed to directly test whether fast-growing species (*S. pacifica*) facilitate the establishment and growth of other species or if *S. pacifica* monocultures persist, potentially excluding other species.

The amelioration of stress by more rapidly colonizing marsh plants can facilitate the colonization of marsh consumers, such as crabs (Daleo et al. 2011; He & Cui 2015). Prior to the planting experiments, we observed more crab (*Uca crenulata*, *Pachygrapsus crassipes*) burrowing activity in vegetated versus adjacent nonvegetated substrate (Schroeter & Page pers. obs.), and have noted a similar pattern following planting. Although plant diversity may be temporarily lower at San Dieguito, the burrowing and foraging activities of crabs can improve overall marsh functioning (Escapa et al. 2008; Daleo et al. 2011; Beheshti et al. 2022). Such plant–animal interactions may be mutualistic (Bertness 1985). For example, plants can facilitate crab expansion into higher elevation areas (He & Cui 2015), which, in turn, creates the topographic heterogeneity often required for the successful establishment of marsh plants (Brooks et al. 2015).

The importance of freshwater events in the functioning of salt marshes has been well-documented in Mediterranean

(Allison 1996) and other semiarid environments (Zedler 1983; Zedler & Beare 1986; Noe & Zedler 2001). Temporary alleviation of stressors (hypersalinity, desiccation, low soil moisture) with freshwater inputs can increase the recruitment, germination, and subsequent growth of mid to high salt marsh plants (Zedler 1983; Zedler & Beare 1986; Noe & Zedler 2001), especially *Salicornia pacifica*, (Allison 1996; Noe & Zedler 2001). During periods of drought, irrigation can ameliorate the stress of hypersaline soils and increase germination (Noe 2002; Xie et al. 2019a), potentially creating safe sites for plant establishment (Harper 1977). Our findings are consistent with previous studies in natural wetlands where irrigation of bare patches in the high marsh temporarily reduced soil salinities and facilitated the emergence of seedlings (Shumway & Bertness 1992; Noe & Zedler 2000, 2001). In our study, alleviating key stressors (hypersalinity, compact soils) with irrigation and decompaction enabled marsh plants to establish, persist, and expand throughout the study period following the initial intervention. Our results indicate that intervention was necessary to jump-start the establishment and growth of marsh plants that are now self-sustaining.

In contrast to prior work that showed a positive effect of soil amendment (kelp) on planting survival and growth but no effect of decompaction (O'Brien & Zedler 2006), we found decompaction in combination with irrigation increased vegetation cover, irrespective of soil amendment. Marsh vegetation at San Dieguito is not likely nutrient-limited (Page & Schroeter 2022, University of California, Santa Barbara, Santa Barbara, CA, personal communication), which may explain the lack of a soil amendment effect on vegetation cover. In contrast, amending marsh soils at a nearby restored wetland in the Tijuana estuary found elevated inorganic nitrogen and Total Kjeldahl Nitrogen (TKN) relative to that of reference wetlands (O'Brien & Zedler 2006). In our study, irrigation may have ameliorated the stress associated with compaction (hypersalinity, low soil moisture, and poor drainage), highlighting the importance of using irrigation and decompaction in combination. These contrasting results from neighboring wetlands highlights the value of site-specific studies that identify the particular stressors inhibiting plant colonization post-restoration.

The application of seed mix was not successful at increasing vegetation cover in the mid (Fill In) or high (IDAPS) marsh. In the IDAPS experiment, cover was low in seeded plots and high in not seeded plots that were irrigated and decompacted. In the Fill In, vegetation cover was low in seeded and control plots (<10%). Out of concern that the hand-broadcast seeds would drift away with the tides, seeds were raked in and pressed into the marsh soil producing a smooth, flat sediment surface. Introducing topographic heterogeneity can promote seedling establishment and growth of salt marsh species by reducing tidal flow velocity (Fivash et al. 2021), increasing oxygen availability, improving drainage (Fivash et al. 2020), and enhancing seed retention (Balke et al. 2014, 2016). The lack of topographic heterogeneity in seeded plots in the IDAPS and Fill In experiments may have contributed to their low cover. Further, tidal flow during inundation may inhibit the retention of seeds on the marsh plain and the “window of opportunity” that seedlings require

to firmly attach to the substrate and survive (Balke et al. 2014; Hu et al. 2015). Seedlings are particularly vulnerable in the bare or sparsely vegetated landscape of San Dieguito's mid to high marsh where there is little protection from dislodgement and desiccation stress, particularly in such semiarid environments (Balke et al. 2014). Restoration projects that attempt to revegetate via seeding are likely to be most successful in areas with sufficient drainage (minimize ponding) and long inundation periods where conditions are more likely to lead to successful germination, and involve methods (burlap; Zedler et al. 2003) that secure seeds from drifting away with the tides without compromising topographic heterogeneity.

The Value of Field Experiments that Inform Best Practices for Establishing Vegetation in Stressful Environments

Field experiments of sufficient duration capture variability through space and time and allow time for delayed treatment effects to emerge (Witman et al. 2015; Beheshti et al. 2021), such as those observed in the IDAPS experiment. Our results illustrate the challenges of developing vegetation cover at mid to high marsh elevations and the value of longer-term, iterative, in-situ investigations for determining best practices to achieve vegetation targets. The recovery response of wetland vegetation to disturbance can be variable, but is often delayed as evidenced by well-documented time lags in marsh response to changing conditions (experimental or natural) in perennial salt marsh systems (Lohmus et al. 2019; Beheshti et al. 2021, 2022). These conditions require monitoring experiments over extended time frames that can capture this variability across multiple growing seasons (Zedler et al. 2003; Nolan et al. 2020; Rabinowitz et al. 2022).

Future studies that measure both soil properties and vegetation cover are needed to track potential changes in edaphic conditions in response to applied treatments in order to more directly relate those changes to planting survivorship, seedling establishment, and plant growth rate. Our experiments were not designed to evaluate the effects of soil properties on vegetation cover. Consequently, additional soil data are needed to confirm our assumption that irrigation and decompaction decreased soil salinity and increased drainage, facilitating *S. pacifica* recruitment and expansion.

Practical Application of Adaptive Restoration

Although the number of wetland restorations is increasing, the transferability of lessons learned from project-to-project is a challenge, given site-specific differences. Thus, we are largely still in the "proof-of-concept" stage of wetland restoration science (Suding 2011; Bayraktarov et al. 2020). Furthermore, the rapidity with which positive wetland restoration outcomes are realized is highly variable (Zedler & Kercher 2005; Moreno-Mateos et al. 2012). Implementing a well-designed wetland restoration does not necessarily confer project "success" (Cadier et al. 2020) or progress towards project targets (Zedler & Callaway 1999; Zedler 2007). Site-specific investigations are important for identifying the mechanisms of restoration underperformance or unexpectedly slow recovery trajectories.

Our investigations follow the adaptive restoration approach prescribed by Zedler (2017). We generated data that directly informed management decisions to help the restored wetland reach a performance target for vegetation cover defined by the cover of reference wetlands in the region. Given the site-specific challenges of establishing marsh vegetation, we recommend others follow a similar approach of (1) identifying leading stressors that may inhibit recovery, (2) establishing, through controlled field experiments, a mechanistic linkage between plant performance and identified stressors, (3) developing best practices for active restoration (planting, seeding) to hasten recovery, and (4) using the results of these mechanistic field experiments to apply a science-based adaptive restoration approach.

Acknowledgments

We thank the staff of the San Onofre Nuclear Generating Station (SONGS) Marine Mitigation Program, Marine Science Institute, University of California, Santa Barbara, for helping to collect and process the data—E. Blair, C. Haack, D. Huang, I. Scott, M. Tusa, and D. Weisman. We also like to thank Drs. Peter Raimondi, Richard Ambrose, and Russell Schmidt for stimulating discussions on all aspects of this study. Funding was provided by Southern California Edison (SCE) as required by the California Coastal Commission under SCE's coastal development permit (No. 6-81-330-A, formerly 183-73) for Units 2 and 3 of the San Onofre Nuclear Generating Station. Data are available on the Environmental Data Initiative (EDI) Data Portal <https://doi.org/10.6073/pasta/a3d22413d740799af91924673b8fb556>; <https://doi.org/10.6073/pasta/4bf655237e7e71f7e45cd28fe0afcef3>; <https://doi.org/10.6073/pasta/07a70b32d7e398b9d95f52aab14f31ae>.

LITERATURE CITED

- Allison SK (1996) Recruitment and establishment of salt marsh plants following disturbance by flooding. *The American Midland Naturalist* 136:232–247. <https://doi.org/10.2307/2426728>
- Balke T, Herman PMJ, Bouma TJ (2014) Critical transitions in disturbance-driven ecosystems: identifying windows of opportunity for recovery. *Ecology* 102:700–708. <https://doi.org/10.1111/1365-2745.12241>
- Bayraktarov E, Brisbane S, Hagger V, Smith CS, Wilson KA, Lovelock CE, Gillies C, Steven ADL, Saunders MI (2020) Priorities and motivations of marine coastal restoration research. *Frontiers in Marine Science* 7:484. <https://doi.org/10.3389/fmars.2020.00484>
- Beheshti K, Endris C, Goodwin P, Pavlak A, Wasson K (2022) Burrowing crabs and physical factors hasten marsh recovery at panne edges. *PLoS One* 17: e0249330. <https://doi.org/10.1371/journal.pone.0249330>
- Beheshti K, Wasson K, Angelini C, Silliman BR, Hughes BB (2021) Long-term study reveals top-down effect of crabs on a California salt marsh. *Ecosphere* 12:e03703. <https://doi.org/10.1002/ecs2.3703>
- Bertness MD, Callaway R (1994) Positive interactions in communities. *Trends in Ecology and Evolution* 9:191–193. [https://doi.org/10.1016/0169-5347\(94\)90088-4](https://doi.org/10.1016/0169-5347(94)90088-4)
- Bertness MD (1985). Fiddler crab regulation of spartina alterniflora production on a new england salt marsh. *Ecology* 66(3):1042–1055. <https://doi.org/10.2307/1940564>
- Bertness MD, Ewanchuk PJ, Silliman BR (2002) Anthropogenic modification of New England salt marsh landscapes. *Proceedings of the National Academy of Sciences U.S.A.* 99:1395–1398. <https://doi.org/10.1073/pnas.022447299>

- Bertness MS, Ellison AM (1987) Determinants of pattern in a New England salt marsh plant community. *Ecological Monographs* 57:129–147. <https://doi.org/10.2307/1942621>
- Bouma TJ, De Vries MB, Herman PMJ (2010) Comparing ecosystem engineering efficiency of two plant species with contrasting growth strategies. *Ecology* 91:2696–2704. <https://doi.org/10.1890/09-0690.1>
- Bouma TJ, De Vries MB, Low E, Peralta G, Tanczos C, Van de Koppel J, Herman PMJ (2005) Trade-offs related to ecosystem engineering: a case study on stiffness of emerging macrophytes. *Ecology* 86:2187–2199. <https://doi.org/10.1890/04-1588>
- Boyer KE, Fong P, Vance RR, Ambrose RF (2001) *Salicornia virginica* in a Southern California salt marsh: seasonal patterns and a nutrient-enrichment experiment. *Wetlands* 21:315–326. [https://doi.org/10.1672/0277-5212\(2001\)021\[0315:SVIASC\]2.0.CO;2](https://doi.org/10.1672/0277-5212(2001)021[0315:SVIASC]2.0.CO;2)
- Boyer KE, Zedler JB (1999) Nitrogen addition could shift plant community composition in a restored California salt marsh. *Restoration Ecology* 7:74–85. <https://doi.org/10.1046/j.1526-100X.1999.07109.x>
- Brooks KL, Mossman HL, Chitty JL, Grant A (2015) Limited vegetation development on a created salt marsh associated with over-consolidated sediments and lack of topographic heterogeneity. *Estuaries and Coasts* 38:325–336. <https://doi.org/10.1007/s12237-014-9824-3>
- Brophy LS, Greene CM, Hare VC, Holycross B, Lanier A, Heady WN, O'Connor K, Imaki H, Haddad T, Dana R (2019) Insights into estuary habitat loss in the western United States using a new method for mapping maximum extent of tidal wetlands. *PLoS One* 14:e0218558. <https://doi.org/10.1371/journal.pone.0218558>
- Byers JE, Cuddington K, Jones CG, Talley TS, Hastings A, Lambrinos JG, Crooks JA, Wilson WG (2006) Using ecosystem engineers to restore ecological systems. *Trends in Ecology and Evolution* 21:493–500. <https://doi.org/10.1016/j.tree.2006.06.002>
- Byers SE, Chmura GL (2007) Salt marsh vegetation recovery on the Bay of Fundy. *Estuaries and Coasts* 30:896–877. <http://www.jstor.org/stable/27654722>
- Cadier C, Bayraktarov E, Piccolo R, Adame MF (2020) Indicators of coastal wetlands restoration success: a systematic review. *Frontiers in Marine Science* 7:600220. <https://doi.org/10.3389/fmars.2020.600220>
- Callaway JC (2005) The challenge of restoring functioning salt marsh ecosystems. *Journal of Coastal Research* 40:24–36. <http://www.jstor.org/stable/25736613>
- Callaway JC, Sullivan G, Zedler JB (2003) Species-rich plantings increase biomass and nitrogen accumulation in a wetland restoration experiment. *Ecological Applications* 13:1626–1639. <https://doi.org/10.1890/02-5144>
- Clements FE (1916) Plant succession, an analysis of the development of vegetation. Carnegie Institute, Washington DC. <https://doi.org/10.5962/bhl.title.56234>
- Cribari-Neto F, Zeileis A (2010) Beta regression in R. *Journal of Statistical Software* 34:1–24. <http://doi.org/10.18637/jss.v034.i02>
- Cui B-S, He Q, A Y (2011). Community structure and abiotic determinants of salt marsh plant zonation vary across topographic gradients. *Estuaries and Coasts* 34(3):459–469. <https://doi.org/10.1007/s12237-010-9364-4>
- Daleo P, Alberti J, Iribarne O (2011) Crab herbivory regulates re-colonization of disturbed patches in a southwestern Atlantic salt marsh. *Oikos* 120:842–847. <https://doi.org/10.1111/j.1600-0706.2010.19202.x>
- Davidson NC (2014) How much wetland has the world lost? Long-term and recent trends in global wetland area. *Marine and Freshwater Research* 65:934–941. <https://doi.org/10.1071/MF14173>
- Dobson AP, Bradshaw AD, Baker AJM (1997) Hopes for the future: restoration ecology and conservation biology. *Science* 277:515–522. <https://doi.org/10.1126/science.277.5325.515>
- Elwany MHS (2011) Characteristics, restoration, and enhancement of Southern California lagoons. *Journal of Coastal Research* 59:246–255. <https://doi.org/10.2112/SI59-026.1>
- Escapa M, Perillo GME, Iribarne O (2008) Sediment dynamics modulated by burrowing crab activities in contrasting SW Atlantic intertidal habitats. *Estuarine, Coastal and Shelf Science* 80:365–373. <https://doi.org/10.1016/j.ecss.2008.08.020>
- Fischman HS, Crotty SM, Angelini C (2019) Optimizing coastal restoration with the stress gradient hypothesis. *Proceedings of the Royal Society B: Biological Sciences* 286:20191978. <https://doi.org/10.1098/rspb.2019.1978>
- Fivash GS, Temmink RJM, D'Angelo M, van Dalen J, Lengkeek W, Didden K, et al. (2021) Restoration of biogeomorphic systems by creating windows of opportunity to support natural establishment process. *Ecological Applications* 31:e023333. <https://doi.org/10.1008/eap.2333>
- Fivash GS, van Belzen J, Temmink RJM, Didden K, Lengkeek W, van der Heide J, et al. (2020) Elevated micro-topography boosts growth rates in *Salicornia procumbens* by amplifying a tidally driven oxygen pump: implications for natural recruitment and restoration. *Annals of Botany* 125:353–364. <https://doi.org/10.1093/aob/mcz137>
- Foster-Martinez MR, Variano EA (2018) Biosolids as a marsh restoration amendment. *Ecological Engineering* 117:165–173. <https://doi.org/10.1016/j.ecoleng.2018.02.012>
- Harper JL (1977) Population biology of plants. Academic Press, San Diego
- He Q, Cui B (2015) Multiple mechanisms sustain a plant–animal facilitation on a coastal ecotone. *Scientific Reports* 5:8612. <https://doi.org/10.1038/srep08612>
- Howard RJ, Rafferty PS, Johnson DJ (2020) Plant community establishment in a coastal marsh restored using sediment additions. *Wetlands* 40:877–892. <https://doi.org/10.1007/s13157-019-01217-z>
- Hu Z, van Belzen J, van der Wal D, Balke T, Bing Wang Z, Stive M, Bouma TJ (2015) Windows of opportunity for salt marsh vegetation establishment on bare tidal flats: the importance of temporal and spatial variability in hydrodynamic forcing. *Journal of Geophysical Research Biosciences* 120:1450–1469. <https://doi.org/10.1002/2014JG002870>
- Jeppesen R, Rodriguez M, Rinde J, Haskins J, Hughes B, Mehner L, Wasson K (2018) Effects of hypoxia on fish survival and oyster growth in a highly eutrophic estuary. *Estuaries and Coasts* 41:89–98. <https://doi.org/10.1007/s12237-016-0169-y>
- Jin X, Sun X, Li H, Zhao D, Li D, Wang L, Man X (2021) Changes of plant species diversity and biomass with reclaimed marsh restoration. *Journal of Forestry Research* 32:133–142. <https://doi.org/10.1007/s11676-020-01104-y>
- JMP® (1989–2021) Version 16.0.0. SAS Institute Inc., Cary, NC
- Jones CG, Lawton JH, Shachak M (1994) Organisms as ecosystem engineers. *Oikos* 689:373–386. <https://doi.org/10.2307/3545850>
- Kennish MJ (2002) Environmental threats and environmental future of estuaries. *Environmental Conservation* 29:78–107. <https://doi.org/10.1017/S0376892902000061>
- Kent BJ, Mast JN (2005) Wetland change analysis of San Dieguito Lagoon, California, USA: 1928–1994. *Wetlands* 25:780–787. [https://doi.org/10.1672/0277-5212\(2005\)025\[0780:WCAOSD\]2.0.CO;2](https://doi.org/10.1672/0277-5212(2005)025[0780:WCAOSD]2.0.CO;2)
- Kirwan ML, Megonigal JP (2013) Tidal wetland stability in the face of human impacts and sea-level rise. *Nature* 504:53–60. <https://doi.org/10.1038/nature12856>
- Klute A (1986) Water retention: laboratory methods. Pages 635–662. In: Klute A (ed) *Methods of soil analysis: part 1. Physical and mineralogical methods*. ASA Monograph Number 9, American Society of Agronomy and Soil Science Society of America, Madison, USA
- Lohmus K, Balke T, Kleyer M (2019) Spatial and temporal patterns of initial plant establishment in salt marsh communities. *Journal of Vegetation Science* 31:1122–1132. <https://doi.org/10.1111/jvs.12915>
- Merrifield MS, Hines E, Liu X, Beck MW (2011) Building regional threat-based networks for estuaries in the Western United States. *PLoS One* 6:e17407. <https://doi.org/10.1371/journal.pone.0017407>
- Moreno-Mateos D, Power ME, Comin FA, Yockteng R (2012) Structural and function loss in restored wetland ecosystems. *PLoS Biology* 10:e1001247. <https://doi.org/10.1371/journal.pbio.1001247>
- Noe GB (2002) Temporal variability matters: effects of constant vs. varying moisture and salinity on germination. *Ecological Monographs* 71:427–443. [https://doi.org/10.1890/0012-9615\(2002\)072\[0427:TVMEOC\]2.0.CO;2](https://doi.org/10.1890/0012-9615(2002)072[0427:TVMEOC]2.0.CO;2)
- Noe GB, Zedler JB (2000) Differential effects of four abiotic factors on the germination of salt marsh annuals. *American Journal of Botany* 87:1679–1692. <https://doi.org/10.2307/2656745>

- Noe GB, Zedler JB (2001) Variable rainfall limits the germination of upper intertidal marsh plants in Southern California. *Estuaries* 24:30–40. <https://doi.org/10.2307/1352810>
- Nolan M, Bilodeau C, Stratton L (2020) Trade-offs in restoration: using soil amendments to address restoration challenges for plants. *Ecological Restoration* 38:32–41. <https://doi.org/10.3368/er.38.1.32>
- O'Brien EL, Zedler JB (2006) Accelerating the restoration of vegetation in a southern California salt marsh. *Wetlands Ecology and Management* 14: 269–286. <https://doi.org/10.1007/s11273-005-1480-8>
- Pennings SC, Callaway RM (1992) Salt marsh plant zonation: the relative importance of competition and physical factors. *Ecology* 73:681–690. <https://doi.org/10.2307/1940774>
- Pennings SC, Grant MB, Bertness MD (2005) Plant zonation in low-latitude salt marshes: disentangling the roles of flooding, salinity and competition. *Journal of Ecology* 93:159–167. <https://doi.org/10.1111/j.1365-2745.2004.00959.x>
- R Development Core Team (2022) R: a language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria ISBN 3-900051-07-0. <http://www.R-project.org/> (accessed 15 03 2022)
- Rabinowitz TRM, Lundholm JT, Graham JM, Bowron TM, van Proosdij D (2022) Planting techniques and abiotic variation at two salt marsh restoration sites in the Bay of Fundy. *Restoration Ecology*: 1–11. <https://doi.org/10.1111/rec.13707>
- Shumway SW, Bertness MD (1992) Salt stress limitation of seedling recruitment in a salt marsh plant community. *Oecologia* 92:490–497. <https://doi.org/10.1007/BF00317840>
- Silliman BR, Schrack E, He Q, Cope R, Santoni A, van der Heide T, Jacobi M, van de Koppel J (2015) Facilitation shifts paradigms and can amplify coastal restoration efforts. *Proceedings of the National Academy of Sciences U.S.A.* 45: 14295–14300. <https://doi.org/10.1073/pnas.1515297112>
- Silliman BR, van de Koppel J, Bertness MD, Stanton LE, Mendelssohn IA (2005) Drought, snails, and large-scale die-off of Southern U.S. salt marshes. *Science* 310:1803–1806. <https://doi.org/10.1126/science.1118229>
- Sousa AI, Lillebo AI, Pardal MA, Cacador I (2010) Productivity and nutrient cycling in salt marshes: contribution to ecosystem health. *Estuarine, Coastal, and Shelf Science* 87:640–646. <https://doi.org/10.1016/j.ecss.2010.03.007>
- Suding KN (2011). Toward an era of restoration in ecology: successes, failures, and opportunities ahead. *Annual Review of Ecology, Evolution, and Systematics*, 42(1):465–487. <https://doi.org/10.1146/annurev-ecolsys-1027-10-145115>
- Sullivan MJP, Davy AJ, Grant A, Mossman HL (2017) Is saltmarsh restoration success constrained by matching natural environments or altered success? A test using niche models. *Journal of Applied Ecology* 55:1207–1217. <https://doi.org/10.1111/1365-2664.13033>
- Tanner KE, Wasson K, Parker IM (2022) Competition rather than facilitation affects plant performance across an abiotic stress gradient in a restored California salt marsh. *Restoration Ecology*: 1061–2971. <https://doi.org/10.1111/rec.13746>
- Thomsen AS, Krause K, Appiano M, Tanner KE, Endris C, Haskins J, et al. (2022) Monitoring vegetation dynamics at a tidal marsh restoration site: integrating field methods, remote sensing and modeling. *Estuaries and Coasts* 45:523–538. <https://doi.org/10.1007/s12237-021-00977-4>
- Thorne K, MacDonald G, Guntenspergen G, Ambrose A, Buffington K, Dugger B, et al. (2018) U.S. Pacific coastal wetland resilience and vulnerability to sea-level rise. *Science Advances* 4:eaa03270. <https://doi.org/10.1126/sciadv.aao3270>
- Van Dyke E, Wasson K (2005) Historical ecology of a central California estuary: 150 years of habitat change. *Estuaries* 28:173–189. <https://doi.org/10.1007/BF02732853>
- van Putte N, Temmerman S, Verreydt G, Seutjens P, Maris T, Heyndrickx M, et al. (2020) Groundwater dynamics in a restored tidal marsh are limited by historical soil compaction. *Estuarine, Coastal and Shelf Science* 244: 106101. <https://doi.org/10.1016/j.ecss.2019.02.006>
- Vance RR, Ambrose RF, Anderson SS, MacNeil S, McPherson T, Beers I, Keeney TW (2003) Effects of sewage sludge on the growth of potted salt marsh plants exposed to natural tidal inundation. *Restoration Ecology* 11: 155–167. <https://doi.org/10.1046/j.1526-100X.2003.09983.x>
- Wasson K, Jeppesen R, Endris C, Perry DC, Woolfolk A, Beheshti K, et al. (2017) Eutrophication decreases salt marsh resilience through proliferation of algal mats. *Biological Conservation* 212:1–11. <https://doi.org/10.1016/j.biocon.2017.05.019>
- Watson EB, Wigand C, Oczkowski AJ, Sundberg K, Vendettuoli D, Jayaraman S, Saliba K, Morris JT (2015) Ulva additions alter soil biogeochemistry and negatively impact *Spartina alterniflora* growth. *Marine Ecology Progress Series* 532:59–72. <https://doi.org/10.3354/meps11334>
- Whitcraft CR, Levin LA (2007) Regulation of benthic algal and animal communities by salt marsh plants: impact of shading. *Ecology* 88:904–917. <https://doi.org/10.1890/05-2074>
- Witman JD, Lamb RW, Byrnes JEK (2015) Towards an integration of scale and complexity in marine ecology. *Ecological Monographs* 85:475–504. <https://doi.org/10.1890/14-2265.1>
- Wolters MA, Garbutt A, Bekker RM, Bakker JP, Carey PD (2008) Restoration of salt-marsh vegetation in relation to site suitability, species pool and dispersal traits. *Journal of Applied Ecology* 45:904–912. <https://doi.org/10.1111/j.1365-2664.2008.01453.x>
- Xie T, Cui B, Li S, Bai J (2019b) Topography regulates edaphic suitability for seedling establishment associated with tidal elevation in coastal salt marshes. *Geoderma* 337:1258–1266. <https://doi.org/10.1016/j.geoderma.2018.07.053>
- Xie T, Cui B, Li S, Zhang S (2019a) Management of soil thresholds for seedling emergence to re-establish plant species on bare flats in coastal salt marshes. *Hydrobiologia* 827:51–63. <https://doi.org/10.1007/s10750-018-3589-9>
- Yando ES, Osland MJ, Jones SF, Hester MW (2019) Jump-starting coastal wetland restoration: a comparison of marsh and mangrove foundation species. *Restoration Ecology* 27:1145–1154. <https://doi.org/10.1111/rec.12963>
- Yuan L, Chan YH, Wang H, Cao HB, Zhao ZY, Tang CD, et al. (2020) Windows of opportunity for salt marsh establishment: the importance of salt marsh restoration in the Yangtze Estuary. *Ecosphere* 11:e03180. <https://doi.org/10.1002/ecs2.3180>
- Zedler JB (1983) Freshwater impacts on normally hypersaline marshes. *Estuaries* 6:346–355. <https://doi.org/10.2307/1351393>
- Zedler JB (2007) Success: an unclear, subjective descriptor of restoration outcomes. *Ecological Restoration* 25:162–168. <https://doi.org/10.3368/er.25.3.162>
- Zedler JB (2017) What's new in adaptive management and restoration of coasts and estuaries? *Estuaries and Coasts* 40:1–21. <https://doi.org/10.1007/s12237-016-0162-5>
- Zedler JB, Beare PA (1986) Temporal variability of salt marsh vegetation: the role of low-salinity gaps and environmental stress. *Estuarine Variability* 295–306. <https://doi.org/10.1016/B978-0-12-761890-6.5002-3-X>
- Zedler JB, Callaway JC (1999) Tracking wetland restoration: do mitigation sites follow desired trajectories? *Restoration Ecology* 7:69–73. <https://doi.org/10.1046/j.1526-100X.1999.07108.x>
- Zedler JB, Covin J, Nordby C, Williams P, Boland J (1986) Catastrophic events reveal the dynamic nature of salt-marsh vegetation in Southern California. *Estuaries* 9:75–90. <https://doi.org/10.2307/1352195>
- Zedler JB, Kercher S (2005) Wetland resources: status, trends, ecosystem services, and restorability. *Annual Review of Environmental Resources* 30: 39–74. <https://doi.org/10.1146/annurev.energy.30.050504.144248>
- Zedler JB, Morzaria-Luna H, Ward K (2003) The challenges of restoring vegetation on tidal, hypersaline substrates. *Plant and Soil* 253:259–273. <https://doi.org/10.1023/A:1024599203741>

Supporting Information

The following information may be found in the online version of this article:

Figure S1. Inundation curve and elevation data for the study site.

Figure S2. Results from seeded treatments in IDAPS (A–D) and Fill In (A,D) experiments.

Figure S3. Survey schematic for both IDAPS and Fill In experiments.

Figure S4. Marsh vegetation cover $\pm$ standard error of the mean (SEM) by species in the IDAPS experiment.

Figure S5. Transplant size (cm²) of plants in IDAPS and Fill In experiments over time. Over time, transplants expanded in area, though this pattern was most pronounced and

steady for *Salicornia pacifica* (teal squares) and *Frankenia salina* (blue triangles) plantings in the Fill In (solid lines).

Supplement S1. Evaluation of wetland-wide vegetation cover and planting performance in the IDAPS and Fill In experiments.

Coordinating Editor: Stephen Murphy

Received: 5 November, 2022; First decision: 10 February, 2023; Revised: 29 April, 2023; Accepted: 8 May, 2023